# Max Chan

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**ka1dus.github.io | github.com/ka1dus | Available full-time Jan – Jun 2027**

## EDUCATION

**Nanyang Technological University, Singapore** Aug 2024 – May 2028
Bachelor of Computing in Computer Science

**Tampines Meridian Junior College, Singapore** Jan 2020 – Dec 2021
Singapore-Cambridge GCE A-Levels

## PROJECTS

**Human vs Robot Box-Stacking Game** github.com/ka1dus/so101_stack_game
**Python, LeRobot (ACT), MuJoCo, OpenCV, PyTorch**

- Built a game where a person and an SO-101 arm take turns stacking boxes. The arm's place skill was trained only in a calibrated MuJoCo digital twin on 7,000 IK-generated demonstrations and placed 8 of 10 boxes in real games.
- Wrote a camera-based referee and a safety layer that freezes the arm when a hand enters the workspace. The referee called 75 of 75 turns correctly with zero false losses in 10 simulated games.
- Made the controller run unchanged on the real arm, in sim or on recorded clips, then fixed bugs found in real games and replayed every recorded turn to confirm each fix changed nothing else.

## WORK EXPERIENCE

**Botmind** July 2026 – Dec 2026
**Robot Engineering Intern**

- **Airport surveillance perception.** Ran six detectors (fire, fall, fight, running, crowd density, abandoned baggage) on one GPU feed, combining YOLO, ByteTrack, pose estimation and monocular depth in a threaded runtime where a slow model never blocks the others.
- Exported the models to TensorRT FP16 on an RTX 4070 Ti, cutting latency 1.8–2.4× (up to 365 FPS per model), and validated the engines against PyTorch without labeled data at over 99.6% recall.
- Sent each detection to the robot fleet's REST API as an alert with a snapshot, via a background queue with OAuth auto-renewal and strict credential handling (HTTPS host allowlist, no redirects, validated IDs).
- **Operator troubleshooting assistant.** Built an on-premise RAG chatbot over the robot user guide, fusing BM25 and embedding search with reciprocal rank fusion on a local LLM so no data leaves the network.
- Enforced guardrails in code rather than prompts (citation verification, a grounding gate, a confidence floor, fail-closed hotline escalation) so the bot never answers outside the approved guide.
- **Robot integration.** Work on deployed robots over SSH, adding topic-based joystick control to a ROS 2 PTZ node, updating the socket.io bridge to ROS and setting up a strongSwan IPsec VPN.
- **PPE detection on a robot dog.** Labeled data, trained and field-deployed real-time PPE detection on a quadruped robot's edge device, from concept to a live client demo in an aircraft hangar in under 3 days.

**JIMAICONSULTING PTE LTD** Nov 2025 – Present
**Co-Founder**

- Built an event-driven lead-to-appointment platform where an LLM agent qualifies ad leads over WhatsApp and books them into Google Calendar through tool calls, with automated reminders.
- Designed the data model tracking clients, leads, message states and booking outcomes, with logging and traceability for debugging and reporting.
- Built reusable workflow modules (templated message flows, qualification rules, idempotent triggers and retries) so each new client funnel needed little rework.
- Co-founded the company and led onboarding and support, reaching ~$10,000 revenue in 2 months and 90% client retention after month one.

## SKILLS

**Robotics:** LeRobot, imitation learning (ACT), MuJoCo, ROS 2, inverse kinematics, domain randomization, digital twins
**Vision:** PyTorch, YOLO, OpenCV, TensorRT, CUDA, ONNX, ByteTrack, pose estimation, camera calibration
**Applied AI:** LLM agents, tool calling, RAG, hybrid retrieval, local LLM serving (Ollama, vLLM), FastAPI
**Programming:** Python, C++, Java, SQL, React
**Systems:** Linux, Git, SSH, REST APIs, IPsec VPN (strongSwan)
**Languages:** English, Chinese